

Human Factors in Social Media and the Exploitation of Human Behavior Across Age Demographics

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Abstract

Social media has evolved from a tool made to connect into an engineered system designed to monopolize the user's attention. Driven by the attention economy and its financial incentives, platforms leverage Human Factors Engineering (HFE), as well as cognitive and behavioral psychology, to maximize engagement leading to a rise in social media dependency. This literature review examines the specific design features most effectively used for exploiting human cognition to promote addictive behavior. Through persuasive design, such as the "Hook Model", infinite scroll, and variable reward mechanisms, social media companies take advantage of cognitive vulnerabilities and dopamine responses. This paper further analyzes how these features disproportionately impact the developing brains of children compared to adults. The underdeveloped prefrontal cortexes in adolescents leave them particularly vulnerable to addictive behaviors and result in negative mental health outcomes. This review concludes social media addiction is a strategic choice for tech, calling for reform in interface design and applying HFE principles to prioritize user well-being over profit.

Introduction

Social media has undergone an immense transformation. Once a simple network to connect groups of people, it has evolved into a complex and engaging environment driven by financial models incentivising user attention. Over time, the main objective of these platforms has shifted from connection to exploitation, trying to maximise the time users spend on their devices, treating a user's attention as a profitable commodity (Mujica et al., 2022). Pushing for user retention has led to a widespread rise in social media addiction and significant negative impacts on users' daily lives.

The shift did not happen overnight or by accident; it is the result of human-factors engineering being repurposed to serve corporate incentives. In the realm of social media, the primary objective of human-factors engineering flips from optimizing systems for human safety, to one that leverages how our brains work to maximize profit. Ultimately, human-factors engineering, cognitive science, and behavioral psychology have been used to make our phones as addictive as possible. The relentless drive for user retention has led to a rise in social media addiction and negative impacts on users' mental health (Amirthalingam & Khera, 2024). This literature review will explore what specific features exploit human cognition to cause this addiction, as well as how these designs affect children compared to adults.

Background

In modern years, social media has taken a major shift, moving from a platform to facilitate communication, to one that demands attention hoping to keep users engaged for as long as possible. Many major platforms use principles of human factors engineering and the corresponding psychology to maximize user engagement and keep people on their apps. HFE is traditionally used to make systems safer with many advances in driving and other parts of everyday life. Unfortunately, the advertisement structure for monetization in social media has provided an incentive for tech companies to use these principles for an ulterior motive. By exploiting the way human cognition works, tech companies have been able to grab attention easier and hold it for longer. They work to capitalize on dopamine reward pathways to get and keep users on for as long as possible. Interfaces are specifically designed using frameworks such as the "Hook Model" (Lukyanchikova et al., 2023) to exploit cognitive vulnerabilities. Social media's usage of cognitive ergonomics and design has caused a rise in social media addiction and a decrease of the average attention span. Understanding the mechanisms behind this phenomenon is important when considering the ethical implications of modern interface design. The consequences are even larger when considering the developing brains of children. Their brains are not fully developed and the widespread adoption, even within middle school children, only exacerbates the problem among the younger generation.

Objectives

The main objective of this literature review is to analyze the intersection between human factors engineering and human psychology to get a better understanding of how modern social media design promotes addiction. This review hopes to answer two main questions. First, what are specific UX design features exploiting human behavior the most to maximize attention and therefore social media addiction. The second, is how do these specific design features impact children compared to adults.

Literature Search Procedure

For this review I will be using google scholar to find relevant literature. The goal will be to use the advanced search feature and use key words relevant to the topic such as “Human Factors”, “UI”, “UX”, “social media”, “infinite scroll”, etc. To narrow down the search and find the comparison between social media’s effects on children and adults, demographic search topics will be added such as “adolescent”, and “children”. The target is to find 12-15 peer-reviewed academic papers, prioritizing studies that evaluate user interface features or studies emphasizing the difference between the effects on children and adults.

Table of Papers

Author(s) & Year	Title	Key Findings
Ahmed et al. (2024)	Social media use, mental health and sleep: A systematic review with meta-analyses	Provides evidence of social media use leading to sleep disturbances and mental health issues, particularly in younger populations
Amirthalingam & Khera (2024)	Understanding social media addiction: A deep dive	Identifies social media mechanisms such as infinite scroll and notifications as drivers of addiction and low self-esteem in teens
Clark & Zack (2023)	Engineered highs: Reward variability and frequency as potential prerequisites of behavioural addiction	Explores the biology of addiction explaining how variable reward schedules trigger dopamine responses pushing addiction
David & Roberts (2025)	TikTok brain: An investigation of short-form video use, self-control, and phubbing	Demonstrates short-form videos are more immersive than traditional social media, severely undermining

		self-control and prioritizing screen time over social interaction
De et al. (2025)	Social media algorithms and teen addiction: Neurophysiological impact and ethical considerations	Describes AI algorithms and their exploitation of the vulnerability of adolescents, whose underdeveloped prefrontal cortexes cannot regulate the dopamine surges caused by personalized feeds
Eg et al. (2023)	A scoping review of personalized user experiences on social media: The interplay between algorithms and human factors	Examines algorithmic curation and personalization creating "filter bubbles" to maximize engagement while reducing autonomy and cognitive load
Gashaj (2025)	Analysis of social media addiction in teens and young adults: A recommendation report	Shows high daily screen time causes dopamine imbalances, emotional difficulties and reduced concentration in youth
Larson (2024)	What makes social media so addictive, and how does it affect our mental health?	Explains how combining different UI features manipulates user behavior and degrades mental well-being
Lukyanchikova et al. (2023)	A case study on applications of the Hook Model in software products	Analyzes the intentional human factors engineering in apps like Instagram, specifically how the "Hook Model" (Triggers, Action, Reward, Investment) is used to form habits
Montag et al. (2019)	Addictive features of social media/messenger platforms and freemium games against the background of psychological and economic theories	Connects addictive UI elements to the financial incentives of the attention economy, explaining why platforms are driven to design for extended use
Mujica et al. (2022)	Addiction by design: Some dimensions and challenges of excessive social media use	Argues platforms intentionally hook users by design, mapping user experiences to addiction such as salience, tolerance, and withdrawal

Ra et al. (2018)	Association of digital media use with subsequent symptoms of attention-deficit/hyperactivity disorder among adolescents	Provides data showing high-frequency engagement with digital media is associated with the development of ADHD symptoms in adolescents
Sursely et al. (2025)	Time out: A scoping review of non-duration based social media use measures and adolescent mental health	Argues measuring "time spent" is insufficient, highlights the need to analyze design features and interactive behaviors to understand mental health outcomes in adolescents
Zhao et al. (2016)	The social media ecology: User perceptions, strategies and challenges	Views user behavior and how specific interface designs and affordances dictate user perceptions of social belonging and engagement across different platforms

Results

What features exploit human behavior to cause social media addiction?

To best understand the mechanisms behind social media addiction, it is important to understand that social media addiction is not an accidental phenomenon. It is a byproduct of platforms operating with a financial model relying on user attention. The model incentivizes maximizing the amount of time users spend viewing content/advertisements — colloquially known as the “attention economy”. Because advertisers pay by the view on these platforms, “user attention is the product for which they are paid” (Mujica et al., 2022, p. 2). Platforms do not design for the well-being of their users, rather choosing interfaces with "design elements... prolonging app usage" (Montag et al., 2019, p. 1). By intentionally engineering systems to hoard their users attention in accordance to that incentive structure, these platforms tend to create environments that promote addictive behavior. The platforms make billions, addiction is a side effect.

UI designers working for these companies often leverage a psychological framework known as the hook model. This model is integrated within social media "to engage and retain users through the mechanism of habit formation" (Lukyanchikova et al., 2023, p. 292). The cycle has four distinct steps: Trigger, Action, Variable Reward, Investment. In social media this could present itself as a user being sent a push notification (Trigger), opening the app (Action), seeing a new comment on one of their posts (Variable Reward), and a response to that comment (Investment) setting up the next trigger. By running users through the loop formed by the hook model, social media platforms are able to capitalize on small, initial, conscious interactions and

turn them into time-intensive, cognitive disconnection. Users open the app to respond to a message, and fifteen minutes later, they're still scrolling.

The hook model depends on the power of the variable reward step. More specifically, the ability for the given reward to cause a dopamine response in the user. Social media platforms tend to use methods similar to a slot machine with unpredictable social validation such as likes, comments, and messages causing users to crave more and become addicted. This is because the "neuron firing [is] greatest when the [reward is] variable" (Clark & Zack, 2023, p. 1). A variable reward leads to more dopamine released, and a higher chance for addiction. Social media leverages this social reward uncertainty to create a stronger dopamine response, pushing users to check their devices again and again in pursuit of the next dopamine hit.

To exploit a user's dopamine pathways further, platforms seek to implement a frictionless UI navigation to remove any natural stopping cues that may take the user out of the social media induced trance. Infinite scrolling and autoplay, for example, continuously and seamlessly feed the user content without requiring input from the user. "Since the feed is endless, the user does not think about ending and going to another page or stopping" (Larson, 2024, p. 6). These platforms also employ infinite scrolling as it "makes it simple for users to lose track of time and engage in prolonged platform sessions" (Amirthalingam & Khera, 2024, p. 1). They do away with natural stopping points and create a state of flow within the app that makes disengagement require excess cognitive load.

To aid the effectiveness of the frictionless design within the platform UX, these social media platforms use complex machine learning algorithms to track and learn from user behavior. The algorithms create deeply personalized feeds, prioritizing content that is more likely to capture the individuals interests and emotional triggers. The "personalization [makes the] service more relevant ... curating information of interest" (Eg et al., 2023, p. 1). This builds on the infinite scroll and autoplay by not only bypassing the conscious choice to exit the app, but also hyper-personalizing it to make it significantly harder to resist.

A more recent evolution in social media addictiveness has been the push for shorter-form content. These platforms have realized that shorter videos are more immersive than their longer-form counterparts and image-based feeds. They provide quicker, more continuous visual content which undermines a user's ability to self-regulate. This has presented itself in a way that is seemingly counterintuitive to the concept of social media itself. Once presented as a way for people to connect, short-form content has led to a "greater likelihood of phubbing [phone snubbing] others" (David & Roberts, 2025, p. 1292). Phubbing is the use of a phone in the presence of other individuals, a formerly impolite, anti-social action that has become normalized due to "social" media's effects. The format simply speeds up the dopamine reward cycle, damaging baseline self-control, and capturing attention unlike anything else.

How do those features affect children compared to adults?

The literature shows these persuasive design features impact children and adolescents differently than adults, mostly due to neurological differences. The brain development of adolescents and adults is inherently at a different stage of development. Teens have significantly

less developed prefrontal cortexes that struggle to regulate dopamine levels and surges caused by hyperpersonalized feeds. Adolescents are in a stage where they go through significant brain transformations with "structural shifts in the prefrontal cortex and amygdala". This allows social media's mechanisms to be particularly harmful "because of [the amygdala's] vast effects on emotional control, decision-making, and processing rewards" (De et al., 2025). Since their prefrontal cortexes (the rational decision making part of the brain) is not yet developed, teens are more susceptible to the dopamine surges these platforms cause. The algorithms that "continuously [tailor] feeds to individual preferences" (De et al., 2025) exacerbate that as they are specifically designed to increase screen time and activate the reward centers of the brain. These mechanisms appear to "[accelerate] the development of addictive behaviors" in youth (De et al., 2025).

The effects of addiction are also felt by adolescents as a result of the neurological manipulation social media companies are performing. An increased daily screen time builds a highly destructive cycle with "addictive behavior which is reinforced with social media usage [creating] a dopamine imbalance leading to poor self-perception, concentration, and poor sleeping habits, among other negative impacts" (Gashaj, 2025). This is such a widespread issue that in a study of teens and young adults, the individuals most impacted by social media, "93.8% reported experiencing one or more of the following: depression, anxiety, low self-esteem, body image, sleeping problems, social isolation, and emotional difficulties" (Gashaj, 2025). Meta-analyses confirm these outcomes, finding "significant positive associations between social media use, depression, and anxiety" (Ahmed et al., 2024).

Short-form content has farther reaching impacts than just emotional difficulties for youth. Data reveals a link between the constant digital stimulation children are receiving, and behavioral disorders. A two-year study of 15- and 16-year-olds without significant baseline symptoms found a "statistically significant but modest association between higher frequency of digital media use and subsequent symptoms of ADHD" (Ra et al., 2018). The data showed that "High-frequency engagement in each additional digital media activity at baseline was associated with a significantly higher odds of having symptoms of ADHD across follow-ups" (Ra et al., 2018). This suggests the effects of social media's addictiveness may be fundamentally affecting adolescents and altering their cognitive development over the long term.

While adults have fully formed brains and are less likely to be affected by developmental issues from social media platforms, they are not immune to the effects of persuasive design. In adults, the UX features that these platforms implement undermine daily self-control and lead to antisocial behaviors. I'm sure you've caught yourself scrolling mindlessly at some point. That trend has been pushed by the short-form content platforms like TikTok and Instagram reels prioritize. Showing videos between 30-60 seconds tends to "[undermine] self-control which is associated with increased phubbing behavior" (David & Roberts, 2025). While adults may not face the developmental risks, the effects of addictive, short-form content still severely degrades their ability for self-control and interactions with others.

To truly understand the demographic effects of social media, measuring time spent is insufficient. "Research to understand the role of social media use (SMU) in explaining deteriorating adolescent mental health has been limited by broad, nonspecific measures... specifically 'time spent on social media'" (Sursely et al., 2025). Understanding the demographic impact requires analyzing how different age groups navigate the social media environment and perceive their place on social media. For example, users are constantly thinking about their presence online when sharing and liking media, all of which requires them to consider "'audience' and 'content' when sharing," seeking to "access the people and networks they desire to influence" across various platforms (Zhao et al., 2016). Fully understanding social media addiction will require moving beyond screen time to measure the individual affordances of the UX and how they shape social media interactions across age groups.

Discussion

The literature indicates that social media addiction is not simply a failure of willpower, but rather an outcome of deliberate human factors engineering. Platforms are designed to take advantage of the attention economy model where the primary goal is to monopolize the user's time and attention for financial gain. As Mujica et al. (2022) point out, the reality that a user will compulsively check their phone like a slot machine is "not by accident. [It's] a design technique" that leverages various features to manipulate thousands of years of human evolution. The resulting addiction is exactly what these platforms want as the system combines UX features and manipulates human behavior. Ultimately, users are fighting a system designed to disregard their self-regulation and keep them attached.

The societal fallout of the design choices these platforms make have been widespread. They affect individuals across all age demographics. The drive for continuous engagement and the rise of antisocial habits such as phubbing "has been shown to have a negative impact on relationships among co-workers, friends, family, and romantic partners" (David & Roberts, 2025). For younger users, the consequences are even more severe with high screen time leading to "poor self-perception, concentration, and poor sleeping habits among other negative impacts" (Gashaj, 2025). In general, the human factors engineers behind these apps do their job extremely successfully with the downside of having negative implications on society and the way people interact.

Given the breadth and depth of social media's effects, the ethics of UX design must be more heavily scrutinized. Social media corporations are deploying these persuasive algorithms and designs that affect massive populations without significant/sufficient guardrails. This is especially troubling when it is directed at younger populations. Because teenagers have underdeveloped prefrontal cortexes, they do not have sufficient neurological defenses to regulate the dopamine levels and surges caused by these apps. When the algorithms are designed to maximize engagement and exploit vulnerabilities, that is what they do. As mentioned by Montag et al. (2019), the design elements are exclusively focused on generating the most money for the company, with a blatant disregard for the ethical concerns behind it.

To improve this, the industry must change to human-factors based improvements that prioritize user well-being over maximizing engagement. Considering how difficult it is to get companies to give up money, this will likely have to be connected to a legal framework based on age, and the methods the platforms use to provide users with content. Researchers must analyze specific design features to understand their mental health outcomes at various ages. Designers should abandon the hook model and introduce intentional friction with their apps. Removing autoplay, getting rid of infinite scroll, and limiting any algorithms for underage users would be a good start. Applying human-factors principles to protect the users instead of harming them, can allow these platforms to get back to their roots of connecting people instead of dividing them.

Conclusion

This literature review highlights the fact that social media addiction is the goal of modern social media design. It is not an accident that platforms who profit from a user's attention desire to maintain it. It is obvious that specific features, such as infinite scroll, algorithmic personalization, variable rewards, and short form video, exploit human behavior and dopamine pathways to build habits. While these designs negatively impact adult users, their impacts on children and adolescents is disproportionately large as their youth and adaptability makes them vulnerable to neurological manipulation.

Moving forward, it is vital that these platforms adopt strict ethical design frameworks when developing their interfaces. Prioritizing users over the long-term instead of short-term profits must be at the forefront of design. Given the profound health and social implications, lawmakers will likely need to intervene. Regulating these highly-addictive mechanisms may be necessary, treating them with the same caution as other addictive substances to protect the younger generations as they grow.

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